Meric Altug Gemalmaz, Ph.D.

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EDUCATION

Purdue University, West Lafayette, Indiana

- Ph.D. in Computer Science (Human-AI Interaction Focused) **3.96/4.0** Jan. 2020 – June 2025
- M.S. in Computer Science (Transferred to Ph.D.) **3.93/4.0** Jan. 2019 – Dec. 2019
- B.S. in Computer Science (Software Engineering Track, **Distinction**) **3.92/4.0** Aug. 2015 – Dec. 2018

SKILLS

Programming Languages: Python, C/C++, C#, JavaScript, Java, HTML/CSS, R, SQL, Bash
Machine Learning: Large-scale retrieval and ranking, recommender systems, content-integrity and quality signals, LLM-based signal design and prompting, model evaluation, A/B experimentation and metric design
Machine Learning Tools: scikit-learn, GSL, NumPy, Pandas
Tools & Platforms: Git, Meteor.js, MongoDB, MTurk, Android SDK, Firebase, GDB

EXPERIENCE

ML Research Scientist, Meta, Menlo Park, CA Aug. 2025 – Present

- Contributed to large-scale recommendation and ranking systems, first on content recommendation and content quality for Facebook Groups, now on data and evaluation for AI agent systems.
- Shipped **17** A/B-tested launches in roughly nine months on Groups, across **two** distinct recommendation surfaces and tech stacks: built LLM-based content-quality signals, **two** new candidate generators, and content-integrity detection signals.
- Upheld engineering excellence by ramping onto two tech stacks in my first month, taking production oncall by month four, and always chasing root causes in retrieval and measurement instead of patching symptoms.
- Improve the data AI models learn from for AI agent systems, by building the workflows that create and improve their data, alongside creating evaluations for their decisions.

Web App Development for Data Collection, Purdue Jan. 2020 – May 2025

- Coordinated **1,500+** participants across **4+ MTurk** research projects for data collection and analysis.
- Developed and deployed web applications with **JavaScript** and **Meteor.js**, **MongoDB** back ends, and **HTML**/**CSS** front ends.
- Delivered **90%** participant satisfaction and blocked **80%** of bot attacks via scenario-based, interactive UIs.

Fairness in AI-Driven Gig Work: User-Centric Insights, Purdue Dec. 2023 – May 2025

- Initiated a study with gig workers on the long-term impact of AI-driven assignments at varying fairness levels, yielding insights for fairer algorithmic job management.

Design Implications of Fairness in Human-AI Interactions, Purdue [P1, P3] Mar. 2021 – Dec. 2023

- Conducted **six** human-subject experiments on how loan applicants' repeated interactions with an AI loan-approval system affect fairness perceptions and willingness to keep engaging.
- Simulated **1,000** loan applicant interactions with a **Markov Decision Process** in **Python** to estimate experiment parameters; applied **regression analysis** in **R** to real-world data to uncover human-AI interaction trends.
- Found a critical fairness issue: over **75%** of users continue using biased models out of necessity, not perceived fairness, urging a rethink of usage metrics as fairness indicators.

Data Bias Mitigation Algorithm, Purdue [P2] Mar. 2020 – Mar. 2021

- Used unsupervised learning to detect and mitigate cognitive bias in crowdsourced annotations.
- Applied **probabilistic graphical models** and **Expectation-Maximization** with **GSL** in **C** to reduce annotator bias, improving label accuracy across **100** simulated and **12** real-world tasks.
- Achieved over **10%** higher inferred label accuracy than existing baselines by reducing annotation bias.

HONORS AND AWARDS

Recipient of the Graduate Teaching Award, Purdue Nov. 2022
Recipient of the Raymond Boyce Graduate Teaching Award, Purdue Apr. 2020
Dean's List and Semester Honors (**8 Semesters**), Purdue Aug. 2015 – Dec. 2018

SELECTED PAPERS

[P1] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Fairness Perceptions and Retention in Repeated Interactions with AI-Based Decision Systems." *Proceedings of the 5th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, Oxford, UK, Aug. 2022.

[P2] Meric Altug Gemalmaz, Ming Yin. "Accounting for Confirmation Bias in Crowdsourced Label Aggregation." *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, Online, Aug. 2021.

[P3] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Engagement with and Perceived Fairness of AI Models When Opportunities of Qualification Improvement Exist." *arXiv:2410.03126*